Password Entropy

Password entropy measures the strength of a password based on its length and the variety of characters used. It is expressed in **bits** and represents how unpredictable the password is. The *higher* the entropy, the harder the password is to guess or crack.

We can calculate a password's entropy using the following formula:

$$H = L \times \log_2(C) \quad \textit{or} \quad H = \log_2(C^L)$$

where:

- H = password entropy (measured in bits)
- L = length of the password (number of characters)
- C = number of possible characters (or words) available for each position

This calculation assumes that each character in the password is chosen randomly and independently. In practice, human-generated passwords often have less entropy due to predictable patterns.

For example, a password of length 8 and a character set of 52 characters has entropy:

$$H = 8 \text{ x } \log_2(52) = 45.6 \textit{ bits}$$

How Much Entropy Is Enough?

- **72 bits** is generally seen as a minimum threshold for reasonably strong security today.
- **75-80 bits** is often regarded as "strong" for many use cases.
- **100+ bits** is considered highly secure and is a good target for high-risk scenarios or maximum protection.

For most practical purposes, aiming for **80 bits or more** will providerobust protection against current cracking methods.